

TRACK 01

**Intelligent
Candidate
Discovery**

AI Recruiter Ranker

A CPU-only, explainable ranking system for the Senior AI Engineer JD. It looks for shipped retrieval and ranking work, not just AI keywords.

100K

candidate profiles
streamed

**No network. No GPU.
No hosted LLM calls
during ranking.**

What the JD actually rewards

- **Production retrieval** Embeddings, vector search, hybrid retrieval, ranking systems deployed to users.
- **Senior shipper profile** 5-9 year sweet spot, hands-on Python, product-company delivery, not pure research.
- **Evaluation maturity** NDCG, MRR, A/B tests, offline benchmarks, feedback loops.
- **Recruitability** Recent activity, recruiter response, notice period, location or relocation fit.
- **Anti-keyword stance** Business titles with stuffed AI skills should not outrank real builders.

Hybrid scoring pipeline

1

Parse

Stream JSONL into profile, career, skills, signals.

2

Map

Convert JD needs into title, career, skill, product, availability.

3

Score

Combine weighted evidence with deterministic tie-breaks.

4

Guard

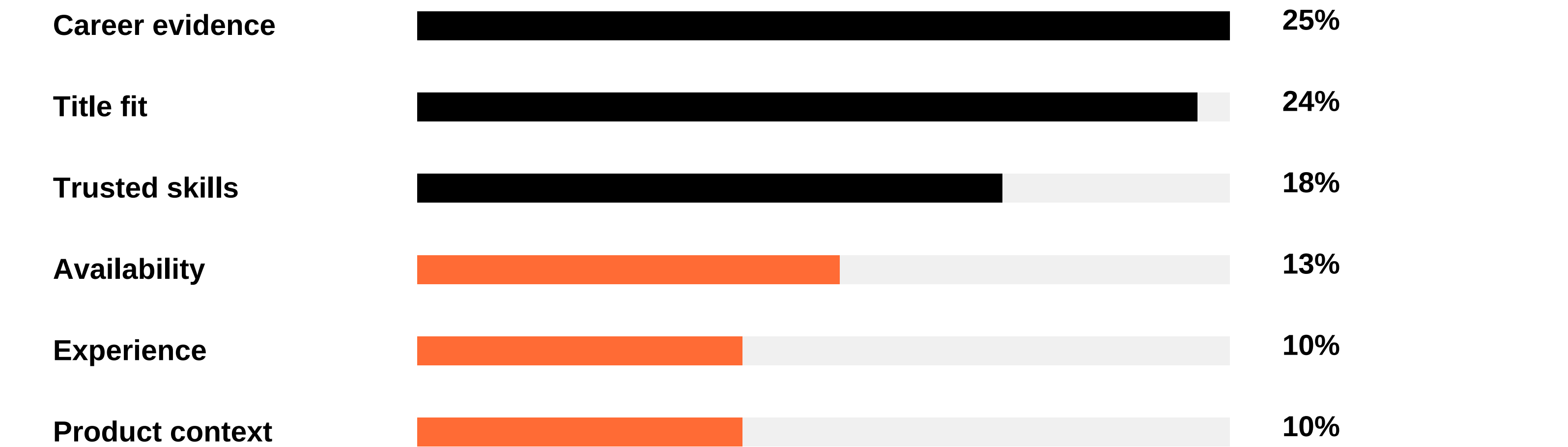
Penalize keyword stuffing and impossible profiles.

5

Explain

Emit top 100 CSV with fact-grounded reasoning.

Signal weights are recruiter-shaped



The largest weights sit on career and title evidence because the JD warns that skills-only matching is the trap.

Submission-ready and reproducible

100

ranked candidates

Valid

official validator

~37s

100K streamed pass

0

runtime dependencies

Top output examples

1. Lead AI Engineer, 6.7 years, retrieval + vector search + recommendations
2. Staff Machine Learning Engineer, 7.0 years, retrieval + semantic search
3. Senior NLP Engineer, 5.2 years, vector search + ranking + embeddings

Reproduce

```
python rank.py --candidates  
./data/candidates.jsonl --out  
./outputs/team_redrob_ranker.csv
```